



Kabul Polytechnic University
Computer Science Faculty
Information System Department

Database II

Lecture 0: Course Overview

10 August – 2026

Contents

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Course Information

- ▶ **Course Name:**

- ✓ Database II

- ▶ **Pre-Requirements for the participation**

- ✓ *Database I*

- ▶ **Type of Course:**

- ✓ Lecture with supporting weekly exercises to repeat and adapt the lecture contents and Projects.

- ▶ **Slides and Extra Notes:**

- ✓ Soft Version, available here: <https://github.com/mujtabaSultani01/Database-II>

Lecture Issues

- ▶ Lecture Times per Week
 - ✓ Monday 11:20 – 12:55 (Lecture)
 - ✓ Wednesday 09:40 – 11:15 (Lecture)
- ▶ Office hours
 - ✓ Saturday 10:00 – 12:00
 - ✓ Sunday 08:00 – 11:00
- ▶ Private appointment
 - ✓ Contact me through email (Mujtaba.cs01@gmail.com).

Assignments

▶ Weekly basis

▶ Rules

- ✓ The Assignments should be handover **Before the deadline...**
- ✓ You will work on the homework in Small groups
- ✓ There should be no copy and paste
- ✓ The copy and paste homework has zero points
- ✓ **Don't Cheat Yourself, Please!!!!...**

Examination and Grading

► Exams

- ✓ Term-wise Project: 20%
- ✓ Final-term Exam: 60%

► Others

- ✓ Class Activity: 10%
- ✓ Homework: 10%

Class Rules

- ▶ Full attendance
- ▶ Please come on time
- ▶ Turn off your mobile.

Don't disturb your classmate !!!

Course Materials

- ❑ Database Systems: A practical approach to design, implementation, and management. ([Main reference](#))
- ❑ Database Systems: Design, Implementation, & Management, by Carlos Coronel and Steven Morris, 13th Ed. ([Main reference](#))
- ❑ Database Systems Concepts, 7th Edition, by Silberschatz, Korth and Sudarshan. ([Supplementary textbook](#))
- ❑ Microsoft SQL Server 2008 R2. ([Supplementary textbook](#))

Course Contents

- ❑ Introduction to Databases
- ❑ Business intelligence
- ❑ Enhanced ERD
- ❑ Relational Algebra
- ❑ Query Optimization
- ❑ Stored Procedure
- ❑ Stored Functions
- ❑ Triggers
- ❑ Transaction Management
- ❑ Indexing
- ❑ Database Administration & Security
- ❑ Distributed DBMSs & Replication Servers

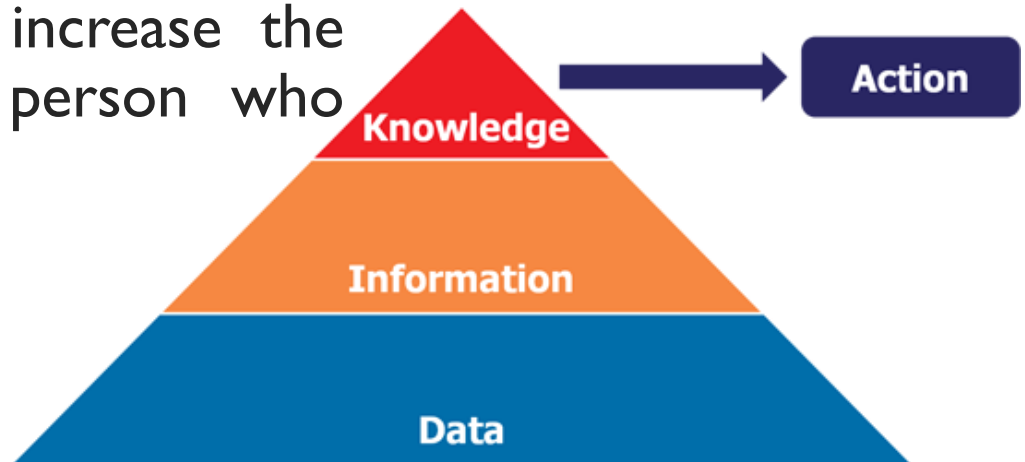
Data & Information

❑ Data

- The term ***data*** referred to facts concerning objects and events that could be recorded and stored on computer media.

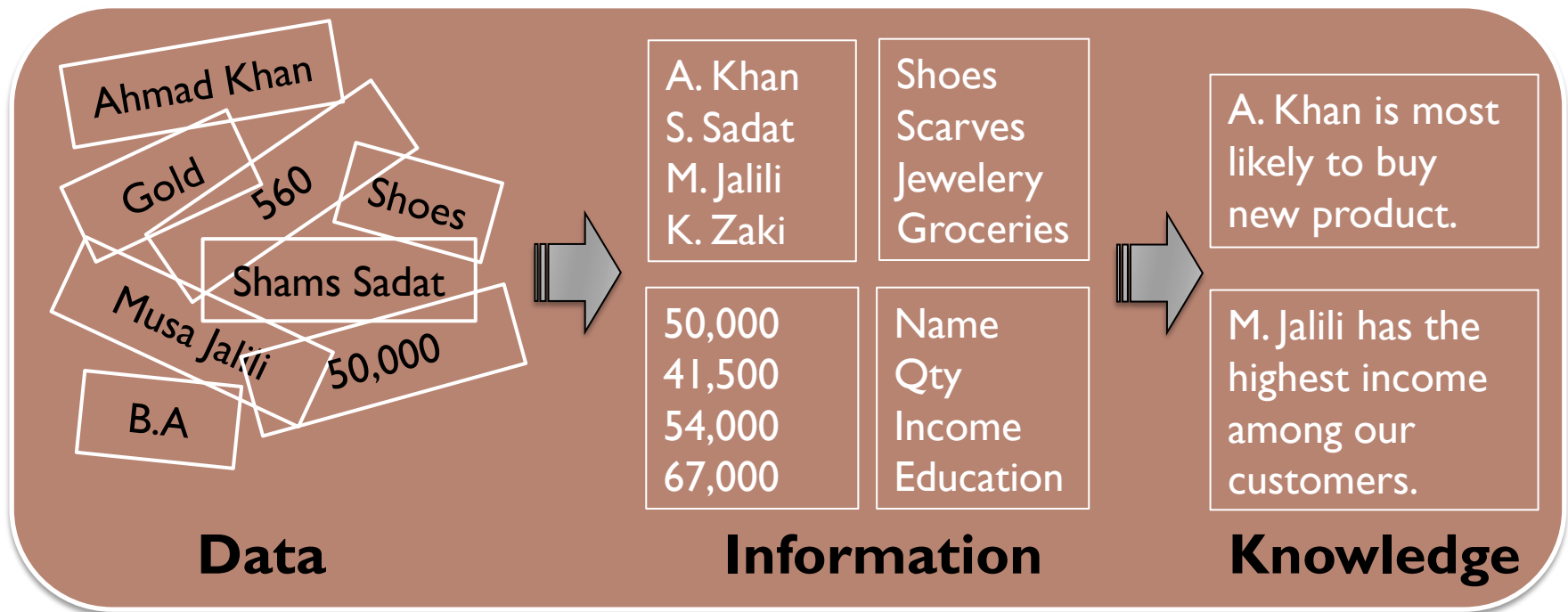
❑ Information

- Data that have been processed in such a way as to increase the knowledge of the person who uses the data.



Data & Information

□ Data, Information & Knowledge



Data Vs Information

Data	Information
Data consist of unprocessed raw facts.	Information the processed form of data.
Data is used as input in the computer.	Used as output of the computer.
Data is not meaningful.	Information is meaningful.
Data normally huge in its volume.	Information normally short in its volume.
Data is difficult to reproduce.	Information easier to reproduced if lost.
Data is un independent entity.	Information is depending on data.
Data is not used in decision-making.	It's very important for decision-making.

Data & Information

❑ Metadata

- Data about the data.
- The metadata describe the data characteristics and the set of relationships that links the data found within the database.

Data & Metadata Examples

Roll No	Name	Address	Email	Phone
1	Ahmad	Karte-Chahar	ahmad.afghan@example.com	+93490004999
2	Husain	Karte-Naw	sayed.husain@example.com	+93917897958

Field Name	Data type	Length	Description	Constraint
Roll No	Integer	3	Roll No of the student	Value from 1 to 100
Name	Alphabetic	50	Name of the student	
Address	Alphanumeric	100	Address of the student	
Email	Alphanumeric	25	Email of the student	Must contain @ and .
Phone	Alphanumeric	25	Phone of the student	
Field Name	Data type	Length	Description	Constraint

File Based System

- ❑ A collection of application programs that perform services for the end-users such as the production of reports. Each program defines and manages its own data.
- ❑ In a typical file processing system, each department in an organization has its own set of files.
- ❑ The files are designed specially for their own application. The records in one file not related to the records in any other file.

File Based System

❑ Problems with File Based System

- Inconsistent Data
- Difficulty of getting quick answers
- Duplication of data
- Data dependence
- Lengthy development times
- Lack of Security
- Excessive program maintenance

File Based System

❑ Example:

Sales Department

Client Details

Name _____
Address _____
Tel.No. _____
Preferred type _____
Max. Rent _____

DreamHome Property for Rent

Address _____
City _____ Postcode _____
Type _____ Rent _____

Owner Details

Name _____
Address _____
Tel.No. _____

Database

- ❑ Organized collection of **logically related data**.
- ❑ Collection of files storing **related data**.
- ❑ Database is an **organized** collection of **related** data that is stored in an **efficient** and **compact** manner.
- ❑ Such as
 - ✓ Library database
 - ✓ University database
 - ✓ Superstore database
 - ✓ Airline reservation database
 - ✓ ...

Introduction to Table

- ❑ Table is the **fundamental object** of the database structure.
- ❑ The basic purpose of table is to **store data**.
- ❑ A table consist of **rows** and **columns**.
- ❑ The data in a table can be **manipulated** easily.

Student ID	Name	Course	Fee
1	Zahir	MOUS	100
2	Nazir	CCNA	200
3	Hilal	MCSE	150
4	Zubair	MySQL	100

What is Row/Record?

- ❑ A logically connected set of one or more fields that describe a person, place or thing is called Row/ Record.
- ❑ Row/Record is the horizontal part of the table.
- ❑ For example the field that comprise a record for student Name Nazir might consist of Nazir's ID, Name, Course and Fee.

Student ID	Name	Course	Fee
1	Zahir	MOUS	100
2	Nazir	CCNA	200
3	Hilal	MCSE	150

What is Column/Field?

- ❑ A column/field is a collection of cells aligned vertically in a table.
- ❑ A column/field is an element in which one piece of information is stored, such as the student's names.
- ❑ Fields/columns are the vertical part of the table.

Name
Zahir
Nazir
Hilal
Zubair

Database Management System

- ❑ **DBMS:** A software system that enables users to define, create, maintain, and control access to the database.
 - Examples: DB2 (IBM), SQL Server (MS), Oracle, Sybase, MySQL, ...
- ❑ **Database System:**
 - The database management system together with the data (database) is called the database system.

Database Management System

❑ **DBMS Functionalities:**

- Defines a particular database in terms of its data types, structures, and constraints.
- Construct or Load the initial database contents on a secondary storage
- Manipulating the database:
 - **Retrieval:** Querying, generating reports
 - **Modification:** Insertions, deletions and updates to its contents.
 - **Accessing** the database through Web applications
- Processing and Sharing
- Protection or Security

Database Advantages

- ☐ Control of data redundancy
- ☐ Data consistency
- ☐ Data sharing
- ☐ Increased productivity of application development
- ☐ Enforcement of standards
- ☐ Improved data quality
- ☐ Improved data accessibility and responsiveness
- ☐ Reduced program maintenance
- ☐ Multi User interfaces

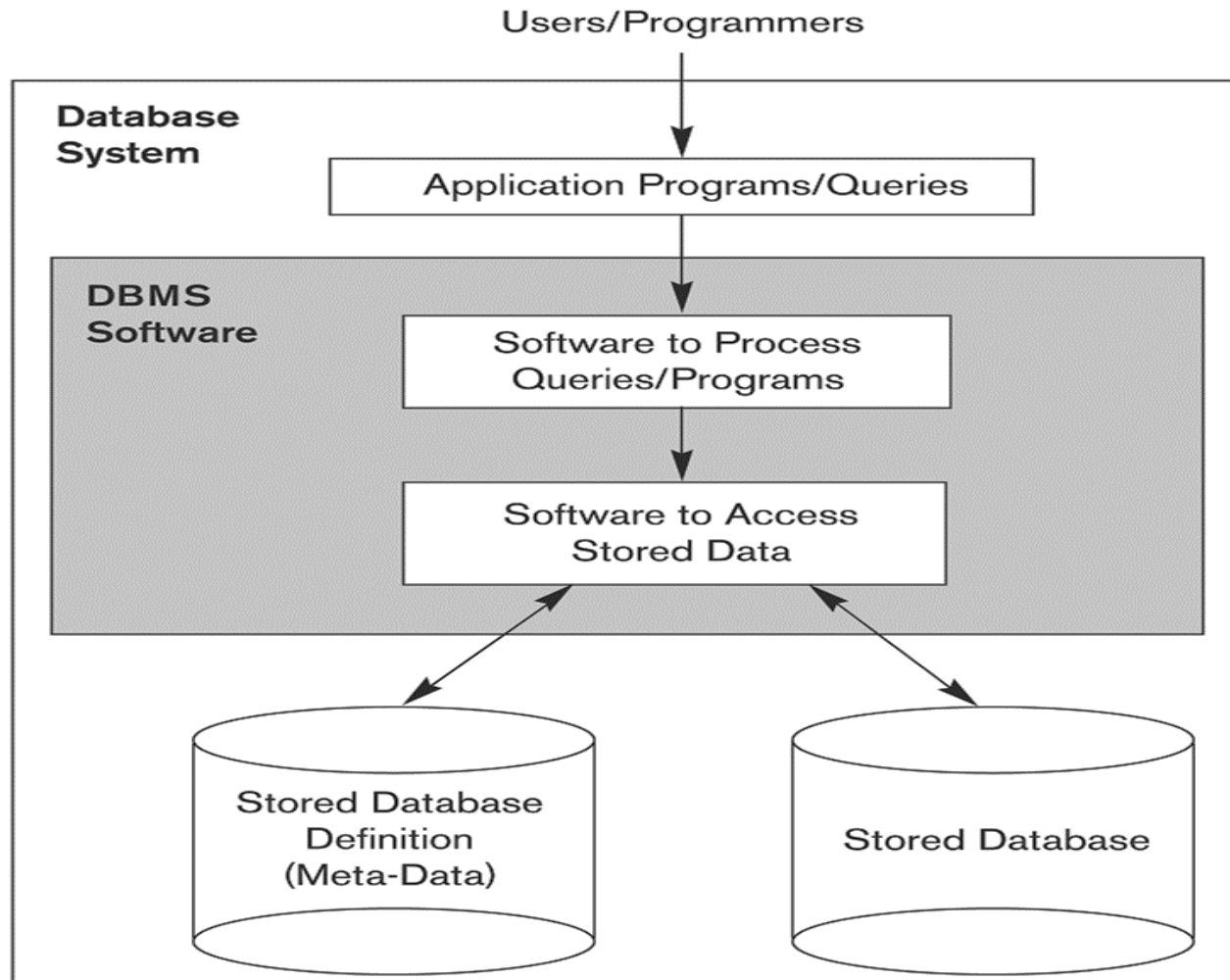


Database Disadvantages

- ☐ Cost of DBMS
- ☐ Complexity (Installation & Management)
- ☐ Hardware Cost
- ☐ Professionals required



Simplified Database Environment



Simple Database Example

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Components of Database Environment

- ❑ **Application Program:** An application program that is used to perform a series of database activities (create, read, update, and delete) on behalf of database users.
- ❑ **Database System**
- ❑ **Database Management System (DBMS)**
- ❑ **Data Repository:** A centralized knowledge base of all data definitions, data relationships, report formats and other system components.
- ❑ **Data Administrator:** Data administrators are persons who are responsible for the overall management of data resources in an organization.

Components of Database Environment

- ❑ **Database Administrators (DBA):** are responsible for physical database design and for managing technical issues in the database environment.
- ❑ **System Developers:** They are systems analysts and programmers who design new application programs.
- ❑ **End-Users:** End users are persons who add, delete and modify data in the database and request information from it.

References

- ❑ Database Systems: A practical approach to design, implementation, and management.

Questions ...?

